

1.5 Benefits of OOP

Following are some advantages of object oriented programming -

1. Using inheritance the ^{unwanted} redundant code can be eliminated and the existing classes can be used.
2. The standard working **modules** can be created using object oriented programming. These modules can then communicate to each other to accomplish certain task.
3. Due to **data hiding** property, important data can be kept away from unauthorized access.
4. It is possible to create **multiple objects** for a given class.
5. For **upgrading the system** from small scale to large scale is possible due to object oriented feature.
6. Due to **data centered nature** of object oriented programming most of the details of the application model can be captured.
7. **Message passing technique** in object oriented programming allows the objects to communicate to the external systems.
8. **Partitioning the code** for simplicity, understanding and debugging is possible due to object oriented and modular approach.

1.7 Applications of OOP

Various applications in which object oriented programming is preferred are -

- Business logic applications
- Knowledge based and expert systems
- Web based applications
- Real time systems
- Simulation and modeling
- Object oriented databases
- Applications in distributed environment
- CAD/CAM systems
- Office automation system
- Games programming

3. Scope Resolution operator

```
#include <iostream>
using namespace std;
int a = 10;
int main()
{
    int a = 200;
    cout << "In value of variable a is" << a;
    return 0;
}
```

Sr. No.	C language	C++ language
1. ✓	C is a procedure oriented language.	C++ is an object oriented programming language.
2. ✓	C makes use of top down approach of problem solving.	C++ makes use of bottom up approach of problem solving.
3.	The input and output is done using <code>scanf</code> and <code>printf</code> statements.	The input and output is done using <code>cin</code> and <code>cout</code> statements.
4. ✓	The I/O operations are supported by <code>stdio.h</code> header file.	The I/O operations are supported by <code>iostream.h</code> header file.
5. ✓	C does not support inheritance, polymorphism, class and object concepts.	C++ supports inheritance, polymorphism, class and object concepts.
6. ✓	The data type specifier or format specifier (<code>%d</code> , <code>%f</code> , <code>%c</code>) is required in <code>printf</code> and <code>scanf</code> functions.	The format specifier is not required in <code>cin</code> and <code>cout</code> functions.

Factorial

```
int fact = 1;  
int fact(int a)
```

```
{  
    fact = fact * a;  
}
```

```
while (i > 0)
```

```
{  
    fact = fact * i;  
    i--;
```

```
}
```

Object is particular entity that has
identity as well as particular for
It is instance of a class

class is collection of object